

SURFACES

VS Code extension

inline hints, navigation

Web dashboard

micro-lessons · quizzes · mastery · practice

Shared editor

instrumented pairing

MICROSERVICES

CODE COACH*detection + hints*

Tree-sitter AST

52 whole-file features

15 error types, 5 ML-gated

3-level scaffolded hints

*also: identity provider***STUDY GUIDER***adaptive remediation*

struggle classifier

(Random Forest)

Graph RAG micro-lessons

auto-generated quizzes

*trigger: 3 repeats per concept***GAMIFICATION***adaptive practice*

mastery-matched practice

progress on resolved bugs

frustration-aware pacing

*consumes shared signals***PAIRPATH***collaboration support*

behavioural events

collaboration-state

classifier (XGBoost)

non-solution nudges

5 states · cooldown policy

DATA

Cloud Firestore

users · diagnostics · events

document store + Neo4j

lessons · mastery graph

progress store

points · levels · streaks

PostgreSQL

sessions · interventions

SHARED CONTRACTS

Identity — one account, issued by Code Coach (Argon2id + rotating JWT); sibling services verify a bearer token via GET /auth/me introspection

Learning events — typed envelope emitted on struggle, hint use and resolution; consumed asynchronously through Pub/Sub push subscriptions